

Quantitative Reasoning

Chapter 3: Ratios & its Applications

Q.1. A jar contains black and white marbles. If there are ten marbles in the jar, then which of the following could not be the ratio of black to white marbles?

- A) 1 : 4
- B) 1 : 10
- C) 7 : 3
- D) 9 : 1

Q.2. Five mangoes and four oranges cost as much as three mangoes and seven oranges. What is the ratio of the cost of one mango to the cost of one orange?

- A) 4 : 3
- B) 1 : 3
- C) 3 : 2
- D) 5 : 2

Q.3. If the income of A is 10% more than that of B and the income of B is 20% less than that of C, then the incomes of A, B and C respectively are in the ratio?

- A) 11 : 10 : 8
- B) 22 : 20 : 25
- C) 10 : 9 : 7
- D) 22 : 18 : 25

Q.4. of 132 examinees of a certain school, the ratio of successful to unsuccessful candidates is 9 : 2. If 4 more students passed, what would have been the ratio of successful to unsuccessful students?

- A) 25 : 4
- B) 4 : 25
- C) 28 : 5
- D) 3 : 28

Q.5. Between two consecutive years my incomes are in the ratio of 2 : 3 and expenses in the ratio of 5 : 9. If my income in the second year is Rs. 45,000 and my expenses in the first year is Rs. 25,000. My total savings for the two years is?

- A) Nil
- B) 5000
- C) 10000
- D) 15000

Q.6. P, Q and R invested Rs. 45000, Rs. 70000 and Rs. 90000 respectively to start a business. At the end of the 2-year period, they earned a profit of Rs. 164000. What will be the Q's share in the profit?

- A) 36000

- B) 56000
- C) 64000
- D) 72000

Q.7. Two friends invested Rs. 1500 and Rs. 2500 in a business. They earned a profit of Rs. 800. One half of the profit was divided equally between them and the other half was divided in proportion to their investment. How much did each of them receive?

- A) Rs. 350 and Rs. 450
- B) Rs. 360 and Rs. 440
- C) Rs. 370 and Rs. 430
- D) Rs. 375 and Rs. 425

Q.8. Shankar started a business with an investment of Rs. 1,20,000. After three months, Aniket joined him with an investment of Rs. 1,90,000. They earned a profit of Rs. 17,50,000 after 1 year. What is Aniket's share in the profit?

- A) 800000
- B) 850000
- C) 900000
- D) 950000

Q.9. Simran started a software business by investing Rs. 50000. After six months, Nanda joined her with a capital of Rs. 80000. After 3 years, they earned a profit of Rs. 24500. What was Simran's share in the profit?

- A) 9423
- B) 10250
- C) 12500
- D) None of these

Q.10. Sonia started a business investing Rs. 60000. After 6 months Vivek joined her with an amount of Rs. 140000. After 1-year Kirti joined them with another Rs. 120000. After 2 years the business yields profit of Rs. 450000. What is the share of Vivek in the profit?

- A) 140000
- B) 198500
- C) 210000
- D) 215000

Q.11. A can finish a work in 18 days and B can do the same work in half the time taken by A. Then, working together, what part of the same work can they finish in a day?

- A) $\frac{1}{6}$
- B) $\frac{1}{9}$
- C) $\frac{2}{5}$
- D) $\frac{2}{7}$

Q.12. George takes 8 hours to copy a 50-page manuscript while Sonia can copy the same manuscript in 6 hours. How many hours would it take them to copy a 100-page manuscript, if they work together?

- A) $6\frac{6}{7}$ hours
- B) 9 hours
- C) $9\frac{5}{7}$ hours
- D) 14 hours

Q.13. Two spinning machines A and B can together produce 3,00,000 metres of cloth in 10 hours. If Machine B alone can produce the same amount of cloth in 15 hours, then what amount of cloth machine A can produce alone in 10 hours?

- A) 50,000 metres
- B) 1,00,000 metres
- C) 1,50,000 metres
- D) 2,00,000 metres

Q.14. A man and a boy together can do a certain amount of digging in 40 days. Their speeds in digging are in the ratio of 8 : 5. How many days will the boy take to complete the work if he engaged alone?

- A) 52 days
- B) 68 days
- C) 80 days
- D) 104 days

Q.15. A group of 12 men can complete a project in 10 days by working 10 hours a day at an efficiency of 80%. How many members are required in a group if the project has to be completed in 8 days with only 60% efficiency and with 8 hours per day shift?

- A) 15 men
- B) 20 men
- C) 25 men
- D) 30 men

Q.16. A car covers a distance of 432 Km at the speed of 48 Km/hr. In how many hours will the car covers this distance?

- A) 6 hours
- B) 7 hours
- C) 9 hours
- D) 12 hours

Q.17. The average speed of a Bus is one-third of the speed of a train. The train covers 1125 km in 15 hours. How much distance will the bus cover in 36 minutes?

- A) 12 Km
- B) 18 Km
- C) 21 Km
- D) None of these

Q.18. A train travels at the speed of 65 Km/hr and halts at 8 Junctions for a certain time. It covers a distance of 1300 Km in 1 day. How long does the train stop at each junction, if it halts for the same period of time at all the stops?

- A) 20 minutes
- B) 30 minutes
- C) 40 minutes
- D) 60 minutes

Q.19. A monkey climbing up the pole ascends 6 meters and slips 3 meter in alternate minutes. If a pole is 60 meters high, how long will it take the monkey to reach the top?

- A) 31 minutes
- B) 33 minutes
- C) 35 minutes
- D) 37 minutes

Q.20. A man in a train notice that he can count 21 telephone posts in one minute. If they are known to be 50 meters apart, then at what speed is the train travelling?

- A) 55 Km/hr
- B) 57 Km/hr
- C) 60 Km/hr
- D) 63 Km/hr

Logical Reasoning

Chapter 3: Seating Arrangement

Q.1. In a gathering seven members are sitting in a row. 'C' is sitting left to 'B' but on the right to 'D'. 'A' is sitting right to 'B'. 'F' is sitting right to 'E' but left to 'D'. 'H' is sitting left to 'E'. Find the person sitting in the middle.

- A) C
- B) D
- C) F
- D) E

Q. 2. A, P, R, X, S and Z are sitting in a row. S and Z are in the centre. A and P are at the ends. R is sitting to the left of A. Who is to the right of P ?

- A) A
- B) X
- C) S
- D) Z

Q.3. P, Q, R, S and T are sitting together. T is at one extreme end. P is the neighbour of T and is third to the left of Q. Who is fourth to the right of T?

- A) P
- B) T
- C) Q
- D) S

Q. 4. There are 8 houses in a line and in each house only one boy lives with the conditions as given below:

- I. Jack is not the neighbour Siman.
- II. Harry is just next to the left of Larry.
- III. There is at least one to the left of Larry.
- IV. Paul lives in one of the two houses in the middle.
- V. Mike lives in between Paul and Larry.

If at least one lives to the right of Robert and Harry is not between Taud and Larry, then which one of the following statements is not correct?

- A) Robert is not at the left end.
- B) Robert is in between Simon and Taud.
- C) Taud is in between Paul and Jack.
- D) There are three persons to the right of Paul.

Q.5. Five girls are sitting on a bench to be photographed. Seema is to the left of Rani and to the right of Bindu. Mary is to the right of Rani. Reeta is between Rani and Mary. Who is sitting at the immediate right of Reeta

- A) Bindu
- B) Rani
- C) Mary
- D) Seema

(Q.6 - Q.8) Eight people- Amar, Ansh, Arun, Amit, Ashu, Anup, Arif and Anil, are sitting around a circular table facing the centre but not necessarily in the same order. Arif is to the immediate left of Ashu. Anup is to the immediate left of Arun. Amar is the neighbour of Ansh. Amit is between Anup and Ansh. Arif is not the neighbour of Ansh or Anil.

Q.6. Who among the following is sitting third to right of Amar?

- A) Arif
- B) Anup
- C) Arun
- D) Ansh

Q.7. How many persons were sitting between Amit and Anil, when counted from the left of Anil?

- A) Less than two
- B) Two
- C) Three
- D) More than three

Q.8. What is the position of Arif with respect to Amit?

- A) Third to the right
- B) Second to the left
- C) Fourth to the right
- D) Fourth to the left

Q.9. Six girls are sitting in a circle facing to the centre of the circle. They are P, Q, R, S, T and V. T is not between Q and S but some other one. P is next to the left of V. R is 4th to the right of P. Which of the following statement is not true?

- A) S is just next to the right to R
- B) T is just next to the right of V
- C) R is second to the left of T
- D) P is second to the right of R

(Q.10-Q.12) Study the given information carefully and answer the questions that follow.

- I. There are seven books, one each of Psychology, Hindi, English, Sociology, Economics, Education and Accountancy lying on a table one above the others.
- II. Sociology is on the top of all books.
- III. Accountancy is immediately below Education which is immediate below Sociology.
- IV. Economics is immediately above Psychology but not in the middle.
- V. Hindi is immediately below Psychology.

Q.10. Economics is between which of the following books?

- A) Accountancy and Education
- B) Psychology and Hindi
- C) English and Psychology
- D) Psychology and Sociology
- E) None of these

Q.11. Which three books are between Accountancy and Hindi?

- A) English, Economics and Psychology
- B) Economics, Psychology and Education
- C) Economics, Psychology and Hindi
- D) Cannot be determined
- E) None of these

Q.12. If Sociology and English, Accountancy and Hindi and Education and Psychology interchange their position, which book will be between Psychology and Sociology ?

- A) Accountancy
- B) Psychology
- C) Hindi
- D) Economics

E) None of these

(Q.13 - Q.15) Read the following information carefully and answer the questions given below it.

Seven friends Kamala, Manish, Rohit, Amit, Gaurav, Pritam and Priya are sitting in a circle. Kamla, Manish, Rohit, Amit, Pritam and Priya are sitting at equal distance from each other. Rohit is sitting two places right of Pritam, who is sitting one place right of Amit. Kamala forms an angle of 90 degree from Gaurav and angle of 120 degree from Manish. Manish is just opposite Priya and is sitting on the left of Gaurav.

Q.13. Who is the only person sitting between Rohit and Manish?

- A) Pritam
- B) Amit
- C) Gaurav
- D) Kamala

Q.14. Gaurav is not sitting at equal distance from which of the pairs

- A) Rohit and Pritam
- B) Amit and Kamala
- C) Manish and Pritam
- D) All of these

Q.15. Gaurav is sitting of Priya.

- A) Sixth to the left
- B) Third to the right
- C) Second to the right
- D) Fourth to the left